

A Novel Label-Guided Attention Method for Multilabel Classification of Multiple Power Quality Disturbances

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Abstract—Multiple power quality disturbance (PQD) contains various single disturbances, so its classification is essentially a multilabel classification task. Due to the complexity of multilabel tasks, the performance of existing multilabel methods for PQD is hard to meet practical needs, which severely restricts the engineering application of multilabel methods. Therefore, in this article, we propose a novel multilabel method named LGAN, which incorporates deep learning and explores the correlations between PQD labels to improve performance. First, 1-D convolutional neural network extracts features automatically from PQD signals. Then, a label-guided attention module is adopted to learn the specific feature representation of each PQD category. Finally, the bidirectional recurrent neural network models the label correlations from the label-related features, and then predicts the final PQD category. Various comparative experiments show that the performance of LGAN is much better than the existing multilabel methods. Additionally, the test using real-time detection platform further verifies the availability of the proposed method.

Index Terms—Deep learning (DL), label correlation, multilabel classification, power quality disturbances (PQD).

I. INTRODUCTION

WITH the massive access of distributed energy systems and various switch devices, power quality disturbances (PQD) occur more frequently [1]. Actually, PQD events are generally not ideal single disturbances, but exist in the form of multiple disturbances in real-life power systems [2]. Compared to single disturbance classification, the difficulties on multiple disturbance classification are mainly manifested in two aspects as follows. 1) The mutual influence of various single disturbances can cause failure features. 2) Too many PQD categories easily lead to the performance degradation of the classifier.

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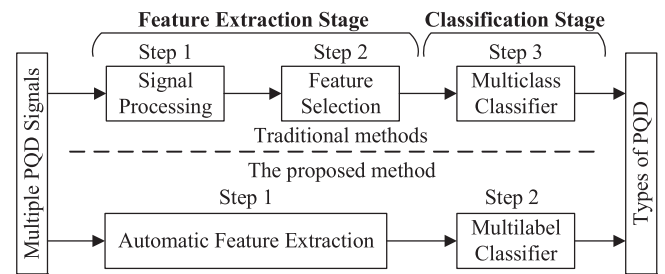


Fig. 1. Framework contrast between the proposed method and the traditional methods.

These difficulties place a higher requirement on the multiple PQD classification methods.

The problem of PQD classification can be divided into two stages: feature extraction and disturbances classification. So far, most studies on multiple PQD classification mainly focus on the improvement of feature extraction techniques. These studies aim at improving feature extraction ability of the model with signal processing algorithms. For example, [3] and [4] extract statistical features, such as energy, mean value, and root mean square, from the coefficients of wavelet decomposition. In the classification stage, these methods carry out PQD classification by using a multiclass approach (regarding each multiple disturbance as a new category). The abovementioned traditional methods extract features in a hand-engineered approach, which have difficulty in selecting effective features. There should be a more efficient method to automatically extract PQD features. In addition, a multiclass approach is still a single-label classification method in essence. The increase of label types leads to the geometric growth of the classification categories, which can reduce the performance of classifiers. To overcome the abovementioned difficulties, we integrate deep learning (DL) with multilabel classifier, and propose a new framework in this article. Compared to traditional methods, the proposed method realizes automatic features extraction by using DL and uses a multilabel classifier instead of a multiclass classifier. The framework contrast between the proposed method and the traditional methods is shown in Fig. 1.

At present, the PQD classification methods that extract features automatically with DL algorithms are well established [5]–[8]. But the existing multilabel classification methods for

PQD classification still have some problems. In fact, each disturbance of multiple PQD is not independent, i.e., the category labels are correlated. In the multilabel problem, using the correlation between labels to improve performance is a challenging task. Luo *et al.* [9] used a conditional probability table to represent the correlation between PQD labels. However, because the conditional probability needs to be extracted from the feature distribution of PQD, this method is not an end-to-end solution. Dong *et al.* [10] used multitask learning to transform the correlations between PQD labels into the correlations between tasks, and proposed an end-to-end multilabel method. But the multitask learning method only considers the correlation between tasks, and ignores the dependencies between different PQD labels in a task.

To address the abovementioned issues, we propose a novel label-guided attention network (LGAN) that incorporates a PQD label to guide learning label-specific features and explores their correlations to improve multilabel classification. More specifically, we first implement a feature extractor based on 1-D convolutional neural network (1DCNN). Then, we design a label-decoupling module that use label-guided attention to learn label-related features. Finally, we utilize a bidirectional recurrent neural network (BRNN) to further model correlations between different PQD labels. The main contributions of this article are as follows.

- 1) We propose a novel multilabel framework for multiple PQD classification, which can automatically extract features and effectively learn the correlations between PQD labels in an end-to-end way.
- 2) Considering the correlation between PQD labels, we use the memory characteristics of RNN to capture the higher order correlations between PQD labels.
- 3) In the process of constructing the network, a label-decoupling module that guides learning label-specific features is designed based on attention mechanism.
- 4) To verify the validity of the proposed LGAN, a large number of comparative experiments with CNN models and other state-of-the-art existing methods for PQD classification are conducted.
- 5) For online real-time detection, we implement the online deployment of the proposed model based on Python and microcontroller (MCU).

The rest of this article is organized as follows. Section II reviews the related work. Section III introduces the design and implementation of LGAN. In Section IV, various experiments and a real-time detection test are conducted. Finally, Section V concludes this article.

II. RELATED WORK

In traditional studies, multiple PQD classification is mainly divided into following three steps: signal processing, feature selection, and classification. These studies aim to obtain better classification performance by improving feature extraction capabilities. The application of signal processing methods, such as Stockwell transform [11], [12], wavelet transform (WT) [3],

[4], empirical mode decomposition (EMD) [13], and variational mode decomposition (VMD) [14], [15], effectively improves the performance of multiple disturbance classification. However, these methods extract features manually, which increases the complexity of the method implementation. Besides, feature selection after feature extraction is also a challenging task. There should be an automatic method to extract PQD features efficiently.

Recently, DL methods have shown excellent feature extraction capability in the field of image, audio, and text classification [16]. Some academics use DL algorithms to extract PQD features and have achieved many advanced results. Mohan *et al.* [5] first explored the performance of various DL methods in the identification of multiple PQD. Liu *et al.* [6], [7] preprocessed the PQD signal with signal processing methods, and then utilized DL algorithms for feature extraction and PQD classification. Wang *et al.* [8] summarized the general design principles and training strategies of DL models in the field of PQD classification. The application of DL methods has greatly improved the feature extraction ability of multiple PQD. Nonetheless, all the abovementioned methods belong to a single-label classification method in essence. The increase of label types leads to the geometric growth of the classification categories, which can reduce the performance of classifiers.

In fact, each multiple disturbance contains various single disturbances, so its classification problem is essentially a multilabel classification problem [17]. Some academics use multilabel classifiers to solve the geometric growth problem of the categories. Zhou *et al.* [17] applied the multilabel classification method based on a ranking approach [18] to the multiple PQD classification problem, and proposes five evaluation metrics for multilabel classification. The proposed K-nearest neighbor Bayesian (KNN-Bayesian) rule is able to identify seven single disturbances and 21 multiple disturbances, which has better classification performance than traditional methods. Guan *et al.* [19] designed a multilabel classification method named ML-RBF. Compared to several common ranking multilabel classification methods, the proposed ML-RBF method has the best classification performance under different noises. The work in [2], [15], and [20] combines multilabel classifiers, such as Rank-SVM and extreme learning machine with signal processing methods and designs a variety of state-of-the-art ranking multilabel classification algorithms. The classification experiments of 47 types of PQD show that the abovementioned methods further improve the multilabel classification performance of multiple PQD. However, these methods do not involve the relative study of label correlations. In [9] and [10], the correlations between PQD labels is considered, but the classification performance needs to be improved.

Different from all these methods, we propose an end-to-end multilabel method, which introduce category information to guide learning label-specific features, and use BRNN to capture the correlations between labels. Consequently, our method can learn the higher order label correlations better and obtain improved classification performance.

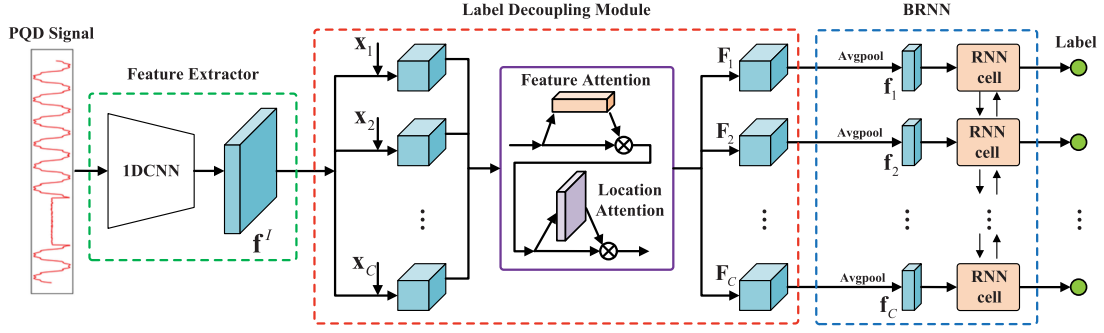


Fig. 2. Framework of the proposed LGAN.

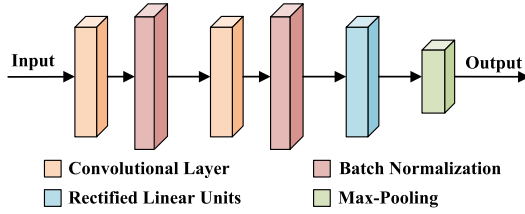


Fig. 3. Basic structure of the convolutional block.

III. DESIGN AND IMPLEMENTATION OF LGAN

A. Framework of LGAN

In this section, an overall description of the proposed LGAN framework is given. Fig. 2 illustrates a detailed pipeline of the LGAN framework. As can be seen from Fig. 2, LGAN mainly consists of three modules: a feature extractor, a label-decoupling module, and a BRNN that characterizes the label correlation.

Given a PQD signal, we first feed it into a deep CNN to generate its feature maps f^I . Then, a label-decoupling module is used to learn the specific features $\{F_1, F_2, \dots, F_C\}$ of each PQD category. The module incorporates the feature vectors $\{x_1, x_2, \dots, x_C\}$ of each PQD label to guide the network to focus on each category label based on feature and location attention. Finally, the BRNN module further captures the correlations between different PQD labels from label-related features, and then predicts the final PQD category label.

B. Feature Extractor

CNN is suitable for extracting the features of array data [21], and PQD signal is essentially a 1-D array data. Therefore, we construct the feature extractor based on 1DCNN. In this article, the feature extractor is composed of stacked convolutional blocks. The basic structure of each convolutional block is shown in Fig. 3.

The PQD signal of n dimension is represented as $x_n = [x_1, x_2, \dots, x_n]$. Then, the process of 1-D convolution of the signal can be expressed as follows:

$$u_i = \sum_{j=1}^m x_{i+j-1} \cdot k_j + b \quad (1)$$

where m is the convolutional kernel size, k_j is the j th weight of the convolutional kernel, and b is the bias term.

Batch normalization (BN) [22] is a normalization processing of input data for each layer during the training process, which can efficiently reduce overfitting and enhance the training speed. BN can be defined as

$$y_i = \gamma \cdot \frac{x_i - \mu_x}{\sqrt{\sigma_x^2 + \epsilon}} + \beta \quad (2)$$

where x_i is the input data, μ_x and σ_x^2 are the mean value and variance of x_i , respectively, and γ and β are the scale and shift parameters, respectively.

The activation function of each convolutional block is rectified linear units (Relu) [23] and expressed as follows:

$$f(x) = \max(0, x). \quad (3)$$

Pooling operation is essentially data dimensionality reduction. Max-pooling is a common pooling method, and expressed as follows:

$$h_{\max} = \max\{x_i | x_i \in \mathbb{N}\} \quad (4)$$

where $\mathbb{N}$ is a set of all data in the area to be pooled.

C. Label-Decoupling Module

After feature extraction, the label-decoupling module is used to generate label-related features. In this article, we utilize a label-guided attention mechanism (including feature and location attention) to implement this module.

Given an input PQD signal I , the feature extractor first extracts its feature maps $f^I \in \mathcal{R}^{L \times N}$, where L and N are the length and channel number of the feature maps, respectively, and formulated as

$$f^I = f_{1\text{dcnn}}(I) \quad (5)$$

where $f_{1\text{dcnn}}(\cdot)$ is a feature extractor.

For each PQD category, a one-hot code vector x_c is used to represent its category information, where c represent the c th PQD category. The length of label vector x_c is the PQD category number C . Then, a low-rank bilinear pooling method [24] is used to fuse the signal features f^I and label features x_c . For each PQD category and location (c, l) , the fusion process can be expressed

as

$$\tilde{\mathbf{f}}_{c,l}^I = \mathbf{P}(\tanh((\mathbf{U}\mathbf{f}_{c,l}^I) \odot (\mathbf{V}\mathbf{x}_c))) + \mathbf{b} \quad (6)$$

where $\tanh(\cdot)$ is the hyperbolic tangent function, $\mathbf{U} \in \mathcal{R}^{d_1 \times N}$, $\mathbf{V} \in \mathcal{R}^{d_1 \times C}$, $\mathbf{P} \in \mathcal{R}^{d_2 \times d_1}$, and $\mathbf{b} \in \mathcal{R}^{d_2}$ are the learnable parameters, which are implemented by using a fully connected layer. $\odot$ is the element-wise multiplication operation, and the d_1 , d_2 , and C are the dimensions of the joint embeddings, output features, and the label vectors, respectively. At this point, an intermediate feature map $\tilde{\mathbf{f}}^I \in \mathcal{R}^{C \times d_2 \times L}$ is obtained.

Then, the channel and spatial attention module proposed in [25] is used to compute attentional coefficients of intermediate feature map $\tilde{\mathbf{f}}^I$ along two separate dimensions: feature d_2 and location L . For each PQD category and feature (c, d) , the feature attentional coefficient $a_{c,d}$ is computed by

$$a_{c,d} = \sigma(\mathbf{W}_1(\mathbf{W}_0(f_{\text{avg}}(\tilde{\mathbf{f}}_{c,d}^I)) + \mathbf{W}_1(\mathbf{W}_0(f_{\text{max}}(\tilde{\mathbf{f}}_{c,d}^I)))) \quad (7)$$

where $\sigma(\cdot)$ is the sigmoid function, and $f_{\text{avg}}(\cdot)$ and $f_{\text{max}}(\cdot)$ denote global average pooling and global max pooling [26], respectively, $\mathbf{W}_0 \in \mathcal{R}^{d_2/r \times d_2}$ and $\mathbf{W}_1 \in \mathcal{R}^{d_2 \times d_2/r}$ are the learnable parameters, which are implemented by using a convolutional layer, and r is the reduction ratio. Note that the Relus activation function is followed by $\mathbf{W}_0$.

For each PQD category and location (c, l) , the location attentional coefficient $b_{c,l}$ is computed by

$$b_{c,l} = \sigma(f_b([f_{\text{avg}}(\tilde{\mathbf{f}}_{c,l}^I); f_{\text{max}}(\tilde{\mathbf{f}}_{c,l}^I)]) \quad (8)$$

where $\sigma(\cdot)$ is the sigmoid function, $f_b(\cdot)$ represents a convolution operation with only one kernel.

Finally, we add feature and location attention to the intermediate feature map $\tilde{\mathbf{f}}^I$ in a sequential way, and can be formulated as

$$\mathbf{F}_c = b_{c,l}(a_{c,d}\tilde{\mathbf{f}}_{c,d,l}^I) \quad (9)$$

where $a_{c,d}$ and $b_{c,l}$ are the feature and location attentional coefficient, respectively, which indicates the importance of feature d and location l . In the end, we obtain the label-related features $\{\mathbf{F}_1, \mathbf{F}_2, \dots, \mathbf{F}_C\}$ for each PQD category.

D. Correlation Between PQD Labels

The BRNN module aims to learn the correlation between PQD labels. Before capturing the label correlation, three convolutional layers and a global average pooling layer are introduced to reduce the dimensionality of label-related features. For each PQD category, the dimensionality reduction process can be expressed as

$$\mathbf{f}_c = f_{\text{avg}}(f_{\text{cnn}}(\mathbf{F}_c)) \quad (10)$$

where $f_{\text{avg}}(\cdot)$ is a global average pooling operation, $f_{\text{cnn}}(\cdot)$ represents the stacked convolution operation, and $\mathbf{f}_c$ is the label-related feature vector. Note that a Relus activation function and a BN layer are followed by each convolution layer.

Once obtaining the feature vectors $\{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_C\}$ corresponding to all PQD categories, a single-layer BRNN is constructed to explore the interactions of these feature vectors. The structure of BRNN is shown in Fig. 4.

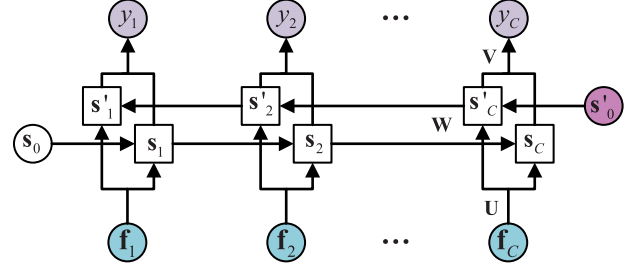


Fig. 4. Basic structure of BRNN.

As can be seen from Fig. 4, each RNN cell consists of two node vectors: forward-state vector s_t and backward-state vector s'_t . The value of each state vector is determined by the input and the previous state, and can be calculated by

$$s_t = \tanh(\mathbf{U}\mathbf{f}_t + \mathbf{W}s_{t-1} + \mathbf{b}) \quad (11)$$

$$s'_t = \tanh(\mathbf{U}'\mathbf{f}_t + \mathbf{W}'s'_{t-1} + \mathbf{b}') \quad (12)$$

where $\tanh(\cdot)$ denotes the hyperbolic tangent function, and $\mathbf{U}, \mathbf{U}' \in \mathcal{R}^{h \times d_2}$, $\mathbf{W}, \mathbf{W}' \in \mathcal{R}^{h \times h}$, and $\mathbf{b}, \mathbf{b}' \in \mathcal{R}^h$ are learnable weights. h and d_2 are the dimensions of state vectors in the hidden layers and feature vectors in the input layers, respectively. Here, the correlations between PQD category labels is modeled as the matrix $\mathbf{W}, \mathbf{W}'$ that describes the relationship between states.

In the output layer, the forward- and backward-state vectors are concatenated into an intermediate-state vector, which is used to predict the confidence score of the presence of the PQD category. In short, the output score is computed as

$$y_t = \sigma(\mathbf{V}([s_t; s'_t]) + b) \quad (13)$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\mathbf{V} \in \mathcal{R}^{1 \times 2h}$ is the weight parameters of linear transformation, and b is the bias term.

Finally, a threshold function $t(\cdot)$ is used to convert the obtained probability $\{y_1, y_2, \dots, y_C\}$ of each PQD category into a binary code, which is the final PQD label. The threshold function $t(\cdot)$ is defined as follows:

$$t(y) = \begin{cases} 1, & y \geq 0.5 \\ 0, & y < 0.5 \end{cases} \quad (14)$$

For each PQD category c , the predicted PQD label is computed as $\tilde{y}_c = t(y_c)$. We perform the process for all categories and obtain a binary label vector $\{\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_C\}$. The "1" in the binary vector means that the input signal contains the corresponding PQD category.

E. Training and Evaluation

In this section, we first give the parameter settings of the proposed LGAN method and the optimization methods in training. Then, we introduce six evaluation metrics and define their expressions.

In this article, we use four convolutional blocks to construct the feature extractor. The number of convolutional kernels in each block is 16, 32, 64, and 128, respectively. Besides, the

kernel size of convolution and Max-pooling is set to 3 and 2, respectively.

For the label-decoupling module, the dimensions d_1 and d_2 of the joint embeddings and output features are all set as 64. N , L , and C are 128, 40, and 8, respectively. In the feature attention module, the reduction ratio r is set as 8. Thus, the kernel numbers of two convolutional layers $\mathbf{W}_0$ and $\mathbf{W}_1$ are 64 and $64/r = 8$. In the location attention module, $f_b(\cdot)$ is implemented by a convolutional layer that consists of only one convolution kernel with the kernel size of 7.

For the BRNN module, the number of neurons h in the hidden layer of BRNN is set as 4. The $\mathbf{V}$ is a 8-to-1 fully connected layer that transforms the state vectors to one single output score.

The cross entropy is adopted as the objective loss function in training. Specifically, given a dataset $\{I_i, \mathbf{y}_i\}_{i=1}^M$ that contains M training samples, in which I_i represents the i th PQD signal, $\mathbf{y}_i = \{y_1^i, y_2^i, \dots, y_C^i\}$ is the corresponding category label. For a PQD signal I_i , we can obtain a predicted score vector $\{\tilde{y}_1^i, \tilde{y}_2^i, \dots, \tilde{y}_C^i\}$. Then, the cross entropy loss can be calculated by

$$\mathcal{L} = -\frac{1}{M} \frac{1}{C} \sum_{i=1}^M \sum_{c=1}^C [y_c^i \log \tilde{y}_c^i + (1 - y_c^i) \log (1 - \tilde{y}_c^i)]. \quad (15)$$

The proposed LGAN is trained with the loss $\mathcal{L}$ in an end-to-end way. The batch size is set to 288 and the number of iterations is 200. The stochastic gradient descent algorithm is the Adam update rule [27], and the initial learning rate is set to 0.001.

The accuracy (ACC) and hamming loss (HL) are used to evaluate the multilabel classification performance of the proposed method. Moreover, to fairly compare with other multilabel methods, we introduced four metrics: ranking loss, one-error, coverage, and average precision (RL, OE, C, and AP, respectively) proposed by [18]. These metrics are defined as follows:

$$\begin{aligned} \text{ACC} &= \frac{1}{n} \sum_{i=1}^n \{f(x_i) = Y_i\}, \quad \text{HL} = \frac{1}{n} \sum_{i=1}^n \frac{1}{q} |f(x_i) \Delta Y_i| \\ \text{RL} &= \frac{1}{n} \sum_{i=1}^n \frac{|D_i|}{|Y_i| |\bar{Y}_i|}, \quad \text{OE} = \frac{1}{n} \sum_{i=1}^n \{f_{\arg\max}(x_i, y_i \notin Y_i)\} \\ \text{C} &= \frac{1}{n} \sum_{i=1}^n \max_{y_j \in Y_i} f_{\text{rank}}(x_i, y_i) - 1 \\ \text{AP} &= \frac{1}{n} \sum_{i=1}^n \frac{1}{|Y_i|} \sum_{y_j \in Y_i} \frac{|L_i|}{f_{\text{rank}}(x_i, y_i)} \end{aligned} \quad (16)$$

where n and q are the numbers of test samples and labels in the labels set, respectively, x_i and Y_i denote the i th samples and its labels set, respectively, and $\bar{Y}_i$ is the complementary set of Y_i . The D_i represents the misjudged label set, $L_i = \{y'_i | f_{\text{rank}}(x_i, y'_i) \leq f_{\text{rank}}(x_i, y_i), y'_i \in Y_i\}$. Δ represents the symmetric difference between two sets, and $f_{\text{rank}}(x_i, y_i)$ returns the rank of y_i . Among these metrics, accuracy and hamming loss are the most important metrics that can evaluate the model more comprehensively.

IV. EXPERIENCE AND RESULTS

According to [2] and standard IEEE-1159 [28], 48 types of PQD signals are simulated in MATLAB. These signals are composed of seven typical single PQD signals (including swell, sag, interruption, flicker, harmonic, impulsive transient, and oscillatory transient). In addition, to ensure that each sample contains at least one PQD category, we also regard the normal signal as a PQD category. Therefore, the sample label is an 8-D vector, in which each binary element indicates the presence or absence of a specific PQD category. The basic frequency f_0 of PQD is 50 Hz, and the sampling frequency f_s is set to 3.2 kHz. Each PQD sample is sampled for ten cycles, with a total of 640 sampling points. For each type of PQD signal, 500 samples under different Gaussian white noise levels are generated. The signal-to-noise ratio (SNR) of each sample is a random value in 20–50 dB. Then, the dataset containing 48×500 samples is evenly divided into training set, validation set, and test set, with a ratio of 3:1:1. In addition, four extra test sets with SNR of 20, 30, 40, and 50 db are generated to evaluate the model.

The experiments are implemented using Python 3.7 environment on a system equipped with an Intel core i7-9700 K 3.6 GHz CPU and a GeForce RTX2070 GPU.

A. Performance Analysis of the Proposed Method

To verify the characteristics and advantages of the proposed LGAN, we establish four advanced CNN models for comparative analysis. The structure and parameters of these methods are as follows.

1) **Multilabel CNN (ML-CNN)**: It is obtained by adding a fully connected neural network including a hidden layer and an output layer after the feature extractor defined in Section III-B. The neuron numbers in the hidden layer and output layer are set to 128 and 8, respectively. Dropout [29] of 0.25 is used between the hidden and output layer to reduce overfitting. The activation function in the output layer is sigmoid.

2) **Multiclass CNN (MC-CNN)**: MC-CNN is exactly the same as the ML-CNN model in structure, but its output layer consists of 48 neurons, and its activation function is softmax.

3) **Deep CNN (DCNN) proposed in [8]**: A deep 1DCNN framework with six convolutional layers and two fully connected layers. The convolution kernel size is 3, and the kernel numbers of each layer are 32, 32, 64, 64, 128, and 128, respectively.

4) **CNN-LSTM proposed in [5]**: A hybrid architecture with two convolutional layers followed by LSTM with 50 memory blocks. The two convolutional layers have 64 and 128 kernels, respectively, with a kernel size of 3.

The abovementioned four CNN models are implemented based on the DL framework Pytorch. Each CNN model is repeatedly trained five times, and the mean $\pm$ standard deviation are taken as the final result. The results are listed in Table I.

As shown in Table I, LAGN achieves the best performance under different SNR. The accuracy of LGAN exceeds 99% except under extreme noise environments (SNR=20 db). Under random noise, the accuracy of ML-CNN, MC-CNN, and DCNN is about 98.5%, and CNN-LSTM obtained the lowest accuracy with 98%. The accuracy gap among ML-CNN and MC-CNN is

TABLE I
DETAILED PERFORMANCE OF DIFFERENT CNN MODELS

Models	Accuracy(%) under different noise levels				
	20dB	30dB	40dB	50dB	rand
ML-CNN	95.92±0.41	98.52±0.21	98.76±0.12	98.96±0.16	98.43±0.11
MC-CNN	96.25±0.29	98.42±0.04	98.54±0.15	98.91±0.08	98.49±0.09
DCNN	96.45±0.24	98.73±0.13	98.76±0.09	99.00±0.09	98.62±0.14
CNN-LSTM	93.87±1.10	98.15±0.48	98.32±0.43	98.09±0.50	98.00±0.48
LGAN	96.85±0.30	99.16±0.08	99.20±0.07	99.30±0.09	99.17±0.15

The test results of the proposed method are shown in boldface.

TABLE II
COMPLEXITY OF DIFFERENT MODELS

Models	FLOPs	Number of parameters	Computational time	
			Training (min)	Testing (s)
ML-CNN	11,107,328	116,568	3.62	14.03
MC-CNN	11,112,448	121,728	7.04	13.91
DCNN	43,580,160	431,504	12.25	29.16
CNN-LSTM	13,895,520	63,408	7.15	24.83
LGAN	28,940,480	199,576	17.12	32.65

The test results of the proposed method are shown in boldface.

very small, which shows that the multilabel classification of PQD cannot significantly improve performance without considering the correlation of the labels.

To analyze the time complexity of the abovementioned models, the parameter quantities and floating point operations per second (FLOPs) of different models are listed in **Table II**.

From the perspective of FLOPs and parameters, the algorithm complexity of the proposed method is relatively high, second only to DCNN in [8]. **Table II** also compares the algorithmic computational time. The testing time of LGAN on a testing set containing 4800 samples is the longest with 32 s. But overall, the test time difference between DCNN and LGAN is of few seconds.

B. Comparative Analysis With Multilabel Methods

To demonstrate the effectiveness of LGAN, three advanced multilabel methods including MLKNN [30], MLHARAM [31], and BPMLL [32] are used for the comparison. Additionally, we combine BPMLL with DL and propose a CNN-BPMLL, which adds the feature extractor defined in Section III-B before BPMLL. The parameters of MLKNN, MLHARAM, and BPMLL are set as the best values reported in previous studies. **Table III** lists the performances of different multilabel methods.

As can be seen from **Table III**, our method obtains the best performance on various metrics under different SNR. The accuracy and hamming loss decrease slightly more only in extreme noise environments (SNR= 20 dB), which illustrates that the antinoise performance of the proposed method is satisfactory. Under random noise, MLKNN and MLHARAM performed the worst with a hamming loss of 10^{-1} , BPMLL followed with a hamming loss of 10^{-2} , and CNN-BPMLL and LGAN were the best with a hamming loss of 10^{-3} . In other words, the hamming loss of LGAN is one to two orders of magnitude lower than that of the other multilabel methods.

TABLE III
PERFORMANCE OF DIFFERENT MULTILABEL METHODS

Metrics	Methods				
	MLKNN	MLHARAM	BPMLL	CNN-BPMLL	LGAN
SNR: 20db					
Hamming Loss↓	0.1160	0.1411	0.0728	0.0079	0.0046
Ranking Loss↓	0.0830	0.1043	0.0215	0.0011	0.0007
One-Error↓	0.0000	0.0000	0.0000	0.0000	0.0000
Coverage↓	3.3012	3.7683	2.6743	2.4311	2.4255
Average Precision↑	0.8169	0.7644	0.9073	0.9977	0.9987
SNR: 30db					
Hamming Loss↓	0.1145	0.1391	0.0647	0.0027	0.0014
Ranking Loss↓	0.0845	0.0929	0.0188	0.0005	0.0003
One-Error↓	0.0000	0.0000	0.0000	0.0000	0.0000
Coverage↓	3.3294	3.6410	2.6427	2.4235	2.4202
Average Precision↑	0.8162	0.7785	0.9164	0.9994	0.9998
SNR: 40db					
Hamming Loss↓	0.1180	0.1296	0.0640	0.0025	0.0013
Ranking Loss↓	0.0857	0.0820	0.0188	0.0004	0.0003
One-Error↓	0.0000	0.0000	0.0000	0.0000	0.0000
Coverage↓	3.3392	3.5038	2.6392	2.4232	2.4201
Average Precision↑	0.8156	0.7952	0.9165	0.9995	0.9997
SNR: 50db					
Hamming Loss↓	0.1172	0.1515	0.0641	0.0022	0.0012
Ranking Loss↓	0.0836	0.1247	0.0185	0.0004	0.0003
One-Error↓	0.0000	0.0000	0.0000	0.0000	0.0000
Coverage↓	3.3073	4.0275	2.6357	2.4226	2.4206
Average Precision↑	0.8164	0.7437	0.9139	0.9995	0.9997
SNR: rand					
Hamming Loss↓	0.1146	0.1474	0.0645	0.0028	0.0013
Ranking Loss↓	0.0823	0.1163	0.0195	0.0004	0.0003
One-Error↓	0.0000	0.0000	0.0001	0.0000	0.0000
Coverage↓	3.2858	3.8987	2.6445	2.4227	2.4208
Average Precision↑	0.8202	0.7200	0.9138	0.9994	0.9997

⁰In the table, ↓ means smaller is better, whereas ↑ means larger is better.

The test results of the proposed method are shown in boldface.

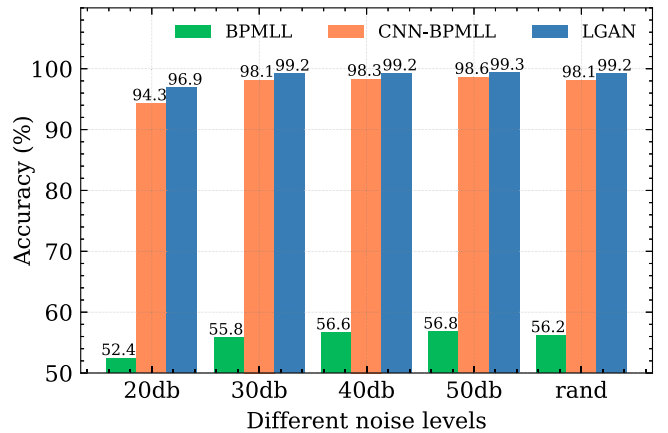


Fig. 5. Accuracy of different multilabel methods.

To further illustrate the improvement of DL and label correlation to PQD classification, we compare the performance of BPMLL, CNN-BPMLL, and LGAN with accuracy as evaluation metric. The comparison result is shown in **Fig. 5**.

Compared to BPMLL, the CNN-BPMLL and LGAN use 1DCNN to extract features, and their performance is significantly superior to that of BPMLL with an accuracy of over 90%. This indicates that DL-based feature extraction methods can effectively improve PQD classification performance. Both

TABLE IV
COMPARISON OF VARIOUS SINGLE-LABEL CLASSIFICATION METHODS

Methods	Feature Extraction	No. of PQD	Accuracy(%)			
			20dB	30dB	40dB	50dB
FDST+DT [12]	Manual	13	-	97.49	98.8	99.28
HST+DT [11]	Manual	13	96.1	97.8	98.7	-
WT+SVM [33]	Manual	9	94.2	-	-	-
FFT+ANN [34]	Manual	8	-	-	96.03	-
DWT+PNN [3]	Manual	16	93.6	95.2	98.6	-
VMD+DT [14]	Manual	9	-	93.8	97.2	-
SSA+CT+DCNN [6]	Auto	22	99.2	-	-	99.35
MFCNN [7]	Auto	24	99.26	-	99.86	-
LGAN	Auto	48	96.85	99.16	99.20	99.30

The test results of the proposed method are shown in boldface.

TABLE V
COMPARISON OF VARIOUS MULTILABEL CLASSIFICATION METHODS

Methods	NO. of PQD	Hamming Loss			
		20dB	30dB	40dB	50dB
KNN+Bayesian [17]	28	0.084	0.083	0.095	0.103
ML-RBF [19]	28	0.069	0.059	0.069	0.08
Rank-WSVM [20]	47	0.152	0.133	0.137	0.139
EEMD+WSVM [2]	47	0.133	0.135	0.1743	0.1714
TL-BN [9]	47	0.1885	0.154	0.1302	0.1202
VMD+RDPEML [15]	47	0.1828	0.1689	0.1723	0.1794
MTL-DN [10]	41	0.0018	0.008	0.014	0.01
LGAN	48	0.0046	0.0014	0.0013	0.0012

The test results of the proposed method are shown in boldface.

CNN-BPMLL and LGAN combine DL with multilabel classification. But different from CNN-BPMLL, the proposed LGAN use label correlations to improve multilabel classification. As can be seen from Fig. 5, the accuracy of LGAN is about 1% higher than that of CNN-BPMLL under various noise levels, which validates the effectiveness of exploring label correlations for multilabel classification of PQD.

C. Comparative Analysis With Other Methods

To further investigate the validity of LGAN, we compare the LGAN with other methods in the existing literatures, and list the comparison results in Tables IV and V.

As shown in Table IV, we compare the existing single-label methods in three aspects: feature extraction method, the number of PQD that can be identified, and accuracy. Compared to single-label classification methods, the proposed LGAN can identify up to 48 disturbances, which greatly increases the number of disturbances that can be identified. On the whole, the methods [6], [7] based on DL have better classification performance than traditional methods. Compared to the existing state-of-the-art DL method [7], the accuracy of our LGAN method is reduced by less than 1%, but the number of PQD categories that can be identified is doubled. The abovementioned results indicate that the proposed LGAN can maintain a high classification accuracy under the premise of expanding the number of disturbance recognition categories.

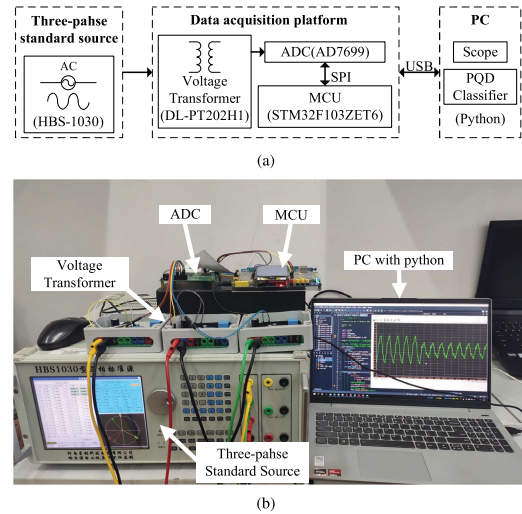


Fig. 6. Real-time detection platform. (a) Framework of the detection platform. (b) Hardware platform.

To investigate the multilabel classification performance of the proposed method, we compare the hamming loss of various existing multilabel classification methods, as listed in Table V. The work in [2], [9], [15], and [20] uses the same simulation dataset as this article, but they were not able to identify normal signals. In terms of hamming loss, the proposed LGAN is two orders of magnitude better than these multilabel methods. The hamming loss of the method in [10] is close to the proposed LGAN, and is only superior to our method at 20 dB SNR. Except [10], the multilabel classification performance of LGAN is one to two orders of magnitude better than the existing multilabel methods. The abovementioned comparison shows the obvious superiority of LGAN for PQD multilabel classification.

D. Real-Time Detection Platform

The LGAN model is deployed on a hardware platform to validate its performance in real-time detection. The real-time detection platform constructs of a three-phase standard source, a data acquisition platform and a PC with Python, as shown in Fig. 6.

Fig. 6(a) shows the framework of the real-time detection platform. The standard source is HBS-1030, which is used to generate seven types of PQD, including C1–C7 (swell, sag, interruption, harmonics, swell with harmonics, sag with harmonics, and interruption with harmonics). The data acquisition platform consists of three parts: voltage transformer, analog-to-digital converter (ADC), and microcontroller (MCU). The current-type voltage transformer DL-PT202H1 steps down the voltage signal generated by the standard source at a transformation ratio of 220 V/4 V. AD7699 is used for data acquisition and conversion. The AD7699 is an 8-channel, 16-bit, charge redistribution successive approximation register ADC, which is capable of converting 500k samples per second. ADC and MCU exchange data with a serial peripheral interface, where the data transfer rate of the AD7699 is set to 32 KHz. The

TABLE VI
PERFORMANCE OF LGAN IN REAL-TIME DETECTION

Type of PQD	C1	C2	C3	C4	C5	C6	C7
Accuracy(%)	100.0	98.33	100.0	98.33	96.67	95.00	96.67
Hamming Loss	0.0	0.0021	0.0	0.0042	0.0042	0.0104	0.0063

MCU STM32F103ZET6 incorporates the ARM Cortex-M3 32-bit core with high-speed embedded memories, where the clock frequency is set to 72 MHz. A PQD real-time detection system is built on the PC with Python. The detection system is responsible for real-time display of signal waveforms and real-time detection of PQD. PC and MCU are connected via serial port, and the sampling rate is set to 3.2 KHz. The hardware platform is shown in Fig. 6(b).

The test signals were generated by manually setting waveform parameters. The detection frequency of the system is set to 5 Hz, i.e., the PQD signal is detected every ten fundamental wave cycles (200 ms). When a PQD signal is detected, the message is displayed on the console and the disturbance waveform is recorded. We repeated the experiment 60 times for each type of disturbance. The real-time detection results are listed in Table VI.

Table VI summarizes the classification performance of the LGAN in real-time detection. The average accuracy is $97.86 \pm 1.72\%$, and the hamming loss is 0.0039 ± 0.0034 . In addition, the recognition and recording time of each PQD in the experiment is 9 ms, which indicates that the recognition speed of LGAN meets the needs of real-time detection. The enhanced classification performance and the real-time detection capability prove the practicability of the proposed LGAN effectively.

V. CONCLUSION

In this article, a multilabel classification method named LGAN based on DL was proposed to recognize multiple PQD. Different from the existing multilabel methods, the proposed LGAN introduces the PQD category information to guide learning label-related features and then capture their correlations to improve the classification performance. In particular, the combination of the proposed label-guided attention module and BRNN can more completely characterize the higher order correlations between PQD category labels. Simulation experiment indicates that the proposed LGAN has high classification performance and good robustness to noise under the premise of greatly expanding the types of PQD that can be identified. Compared to the existing state-of-the-art methods, the LGAN outperforms several powerful multilabel classifiers, such as BPMLL. Furthermore, the real-time detection platform test verify the effectiveness of the method in practical applications.

However, due to the high network complexity of the proposed method, it may be difficult to meet the requirements for delay and computing resources in actual deployment. Further studies should be done with model pruning or knowledge distillation to reduce the computation costs and obtain lower inference time.

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